EXPERIMENT NO:05

CODE:

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Use two features for graph visualization
X = X[:, :2]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Create KNN model
model = KNeighborsClassifier(n_neighbors=5)

# Train model
model.fit(X_train, y_train)
```

```
# Predict output
y_pred = model.predict(X_test)

# Display results
print("Accuracy:", round(accuracy_score(y_test, y_pred) * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Graph 1: Actual vs Predicted classes
plt.figure(figsize=(8, 5))
plt.scatter(
    X_test[:, 0], X_test[:, 1],
    c=y_pred, cmap='viridis', s=80, edgecolors='black'
)
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Sepal Width (cm)")
plt.title("KNN Classification of Iris Flowers")
plt.colorbar(label="Predicted Class")
plt.grid(True)
plt.show()

# Graph 2: Confusion matrix
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 5))
```

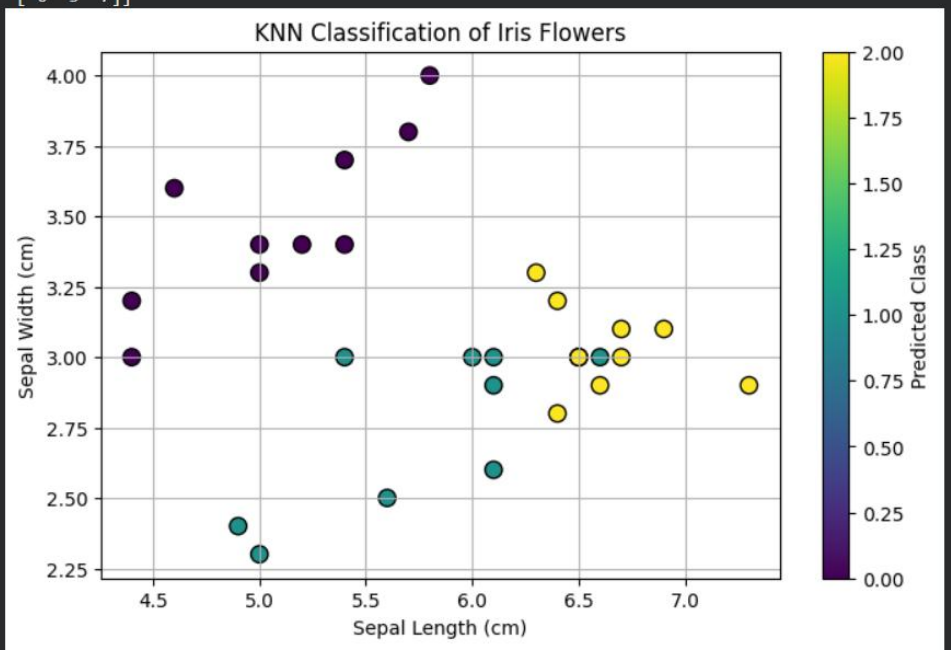
```
plt.imshow(cm, cmap='Blues')
plt.title("KNN Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.xticks([0, 1, 2], iris.target_names)
plt.yticks([0, 1, 2], iris.target_names)
plt.colorbar()

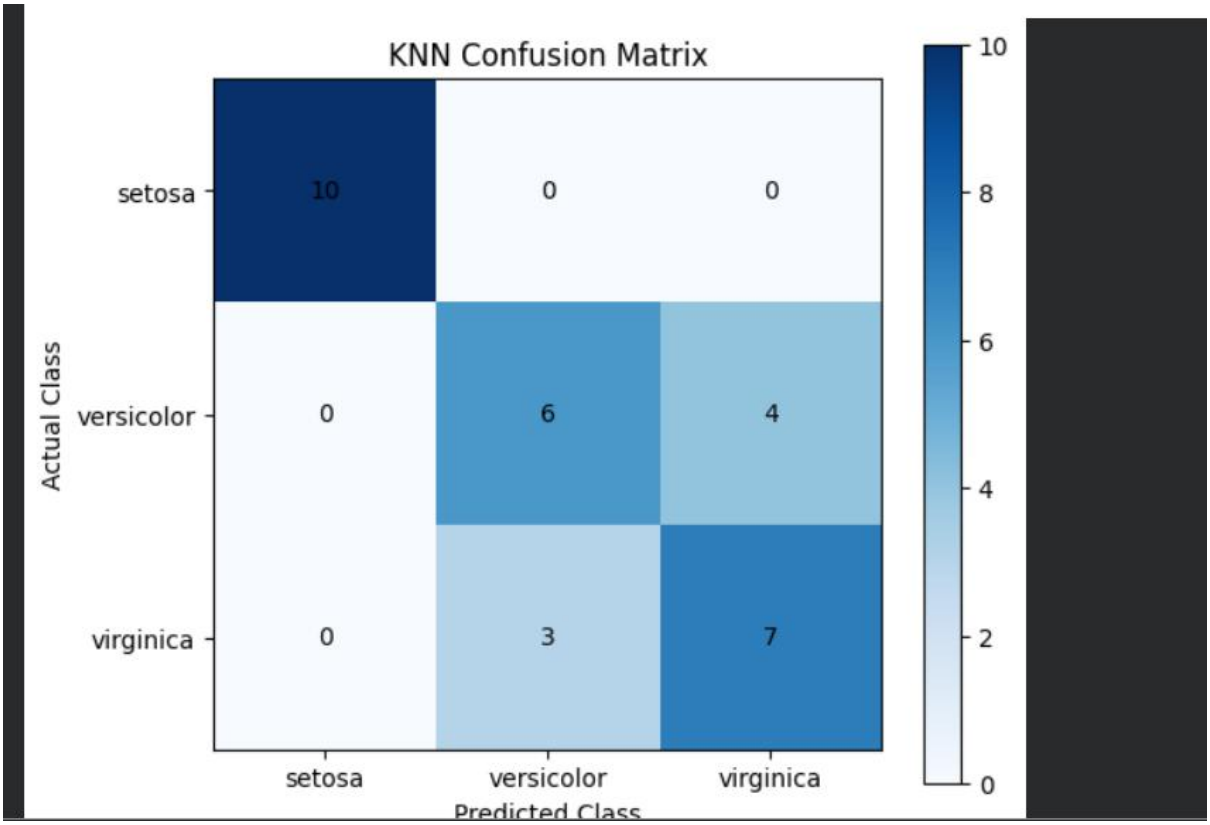
for i in range(3):
    for j in range(3):
        plt.text(j, i, cm[i, j], ha='center', va='center')

plt.tight_layout()
plt.show()
```

OUTPUT:

```
[[10  0  0]
 [ 0  6  4]
 [ 0  3  7]]
```


KNN Classification of Iris Flowers
Sepal Width (cm)
Sepal Length (cm)
Predicted Class
4.00
3.75
3.50
3.25
3.00
2.75
2.50
2.25
4.5
5.0
5.5
6.0
6.5
7.0
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00


KNN Confusion Matrix
Actual Class
setosa
versicolor
virginica
10
0
0
0
6
4
0
3
7
setosa
versicolor
virginica
Predicted Class
10
8
6
4
2
0